Yang(Marino) Li

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Education

Rutgers, The State University of New Jersey-New Brunswick — New Brunswick, NJ, US
Doctor of Philosophy, Computer Science — *Aug 2025 — Present*
- Supervisor: Chengzhi Mao
- Research Interests: Parallel and Spatial Reasoning for (Multimodal) Large Language Models

Hong Kong University of Science and Technology, Guangzhou — Guangzhou, CN
Master of Philosophy, Artificial Intelligence — *Aug 2022 — Oct 2024*
- Supervisor: Ying-Cong Chen, co-supervised by Dan Xu
- Thesis: Neural 3D Reconstruction of Reflective Objects

Sun Yat-sen University — Guangzhou, CN
Bachelor of Science, Mathematics and Applied Mathematics — *Aug 2018 — Jun 2022*
- Thesis: Sim2Real Segmentation for Autonomous Driving

Research Experience

Gemini Team, Google DeepMind — Jun 2026 — Present
Student Researcher — *New York, NY, US*
- **Multimodal Searching Agent Orchestration:** Building a multimodal searching agent system for the Gemini model family.
- Mentor: Guangxin Han

Department of Computer Science, Rutgers University — Aug 2025 — Present
Graduate Research Assistant — *New Brunswick, NJ, US*
- **Parallel Reasoning for Large Language Models:** Conducting research on parallel and interactive reasoning mechanisms. We are developing frameworks that enable LLMs to perform concurrent reasoning with mutual communication, aiming to enhance both inference efficiency and accuracy.
- Mentor: Chengzhi Mao

LightSpeed Studios Research, Tencent IEG — Mar 2025 — Sep 2025
Research Intern — *Shenzhen, CN*
- **Large Generative 3D Models:** Developing a universal 3D asset generation framework, *UltraShape 3D 1.0*, designed to seamlessly integrate with downstream game development pipelines. **Technical report with full-stack open-source is released to public to facilitate community collaboration and innovation.**
- Mentor: Zeyu Hu and Yuhan Wang

Media Computing Group, Microsoft Research Asia — Jun 2024 — Feb 2025
Research Intern — *Beijing, CN*
- **Neural 3D Representation from Unposed Videos:** Proposed an online generalizable 3D Gaussian Splatting (3DGS) reconstruction method for monocular videos. The system transforms video streams into 3D Gaussians within seconds. This work was accepted by **ICCV 2025**.
- Mentor: Jinglu Wang and Xiao Li

Optical Imaging Research Group, SmartMore — Jun 2022 — May 2024
Research Intern — *Shenzhen, CN*
- **Neural 3D Reconstruction with Polarization Cues:** Developed a low-cost and accurate multi-view 3D reconstruction pipeline specifically for reflective objects by leveraging physics-based polarization cues. This work was accepted by **ICLR 2024**.
- Mentor: Jiangbo Lu and Nianjuan Jiang

BME AI Lab, Sun Yat-sen University — Mar 2021 — Nov 2021
Research Assistant — *Guangzhou, CN*
- **Medical Image Segmentation:** Enhanced the accuracy of nasopharyngeal carcinoma segmentation in MRI scans to facilitate precise radiotherapy treatments. The findings were published in the journal *Sensors*.
- Mentor: Zhifan Gao

Publications

Preprints

[Arxiv '26] **Yang Li**, Zirui Zhang, Yang Liu, Chengzhi Mao. LACE: Lattice Attention for Cross-thread Exploration. *arXiv:2604.15529*, 2026.

[Tech Report '25] Tanghui Jia, Dongyu Yan, Dehao Hao, **Yang Li**, Kaiyi Zhang, Xianyi He, Lanjiong Li, Jinnan Chen, Lutao Jiang, Qishen Yin, Long Quan, Ying-Cong Chen, Li Yuan. UltraShape 1.0: High-Fidelity 3D Shape Generation via Scalable Geometric Refinement. *arXiv:2512.21185*, 2025. website

[Arxiv '25] Shiu-hong Kao, Xiao Li, Jinglu Wang, **Yang Li**, Chi-Keung Tang, Yu-Wing Tai, Yan Lu. UVRM: A Scalable 3D Reconstruction Model from Unposed Videos. *arXiv:2501.09347*, 2025. demo

Peer-reviewed

[IROS '26] Yuanjie Lu, Beichen Wang, Zhengqi Wu, **Yang Li**, Xiaomin Lin, Chengzhi Mao, Xuesu Xiao. APPLV: Adaptive Planner Parameter Learning from Vision-Language-Action Model. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.

[ICCV '25] **Yang Li**, Jinglu Wang, Lei Chu, Xiao Li, Shiu-hong Kao, Ying-Cong Chen, Yan Lu. StreamGS: Online Generalizable Gaussian Splatting Reconstruction for Unposed Image Streams. *International Conference on Computer Vision (ICCV)*, 2025.

[ICCVW '25 Oral] Shuai Yang, Yuying Ge, **Yang Li**, Yukang Chen, Yixiao Ge, Ying Shan, Yingcong Chen. SEED-Story: Multimodal Long Story Generation with Large Language Model. *Oral, Workshop on Human-Interactive Generation and Editing, International Conference on Computer Vision (ICCV)*, 2025. code

[ICLR '24] **Yang Li**, Ruizheng Wu, Jiyong Li, Yingcong Chen. GNeRP: Gaussian guided Neural Reconstruction of Reflective Objects with Noisy Polarization Priors. *International Conference on Learning Representations (ICLR)*, 2024. project page

[AAAI '24] Jiyong Li, Dilshod Azizov, **Yang Li**, Shangsong Liang. Contrastive Continual Learning with Importance Sampling and Prototype-Instance Relation Distillation. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2024.

[Sensors '21] **Yang Li**, Guanghui Han, Xiujian Liu. DCNet: Densely Connected Deep Convolutional Encoder Decoder Network for Nasopharyngeal Carcinoma Segmentation. *Sensors 2021, 21(23), 7877*, 2021.

Awards

Ph.D. Fellowship, Department of Computer Science, Rutgers University 2025

Ph.D. Fellowship, Department of Computer Science, Dartmouth College 2025

Stars of Tomorrow (Award of Excellent Intern), Microsoft Research Asia 2025

McGill & Mila Quebec Ph.D. Fellowship, McGill University 2024

Postgraduate Scholarship, HKUST, GZ 2022

Undergraduate Excellent Scholarship, Sun Yat-sen University 2019

Skills

- **Programming Languages**: Python, PyTorch, TensorFlow, MATLAB